Heterogeneous Willingness-to-Pay for Electric Vehicles

Mehrdad Pourghasem

July 23, 2025

Abstract

This paper estimates consumers' willingness-to-pay (WTP) for EVs in the United States, with a focus on how vehicle attributes and household demographics shape adoption decisions. Using a comprehensive market-level data from Ward's Automotive and micro-level household purchase data from the Consumer Expenditure Survey (CEX), I estimate discrete choice demand models—including a random coefficients logit model with demographic interactions and micro-moment conditions. The results reveal substantial heterogeneity in consumer preferences for EVs. While vehicle attributes such as battery capacity and driving range significantly increase utility, a strong baseline disutility for EVs persists, especially among rural, low-income, and non-college-educated households. Urban and college-educated consumers exhibit significantly higher valuations for EVs, and richer households—despite greater purchasing power—demonstrate more price sensitivity, requiring larger incentives to adopt.

1 Introduction

With transportation accounting for nearly a quarter of global carbon emissions (EIA, 2023), transitioning to more sustainable modes of travel is crucial for combating climate change. Electric vehicles (EVs), which produce no tailpipe emissions and are increasingly powered by low-carbon electricity grids, offer a promising strategy to substantially reduce emissions.

Despite extensive policy efforts to promote EV adoption, empirical evidence on how vehicle attributes and consumer demographics shape demand remains limited. Effective policy design requires a nuanced understanding of the sources of consumer resistance and the demographic heterogeneity in willingness to adopt EVs.

This paper estimates U.S. consumers' willingness-to-pay (WTP) for electric vehicles, focusing on how preferences vary with improvements in EV attributes—such as battery capacity and driving range—and across income, education, and geographic segments. To this end, I combine market-level vehicle data from Ward's Automotive (2014–2024) with household-level purchase data from the Consumer Expenditure Survey (CEX). I estimate discrete choice demand models, including a random coefficients logit model with demographic interactions and micro-moment conditions to capture unobserved preference heterogeneity.

Methodologically, the analysis builds on the random coefficients logit model of Berry, Levinsohn, and Pakes (1995), extended to incorporate demographic interactions and micro-moment matching following Petrin (2002). This approach allows the model to fit both aggregate market shares and demographic-specific purchase probabilities.

The results reveal substantial heterogeneity in consumer valuation of EVs. While technical attributes such as battery capacity and driving range increase utility, a large baseline disutility for EVs persists—especially among rural, low-income, and non-college-educated households. In contrast, urban and college-educated consumers exhibit significantly higher valuations for EVs. Richer households, despite greater purchasing power, demonstrate higher price sensitivity, suggesting a need for larger financial incentives to induce adoption.

The existing literature underscores substantial heterogeneity in how consumers value vehicle attributes, particularly fuel economy. Grigolon, Reynaert, and Verboven (2018), for example, analyze new vehicle purchases in Europe from 1998 to 2011, documenting variation in consumer valuation of fuel efficiency across engine types within the same vehicle model. By accounting for anticipated vehicle miles traveled (VMT), they show that consumers expecting to drive more tend to select more fuel-efficient vehicles, suggesting that usage intensity plays

a critical role in attribute-based valuation.

Beyond intrinsic vehicle attributes, recent studies emphasize the role of external market conditions and infrastructure in shaping EV adoption. Li et al. (2017) highlight the indirect network effects of vehicle availability and charger density in influencing adoption behavior. Springel (2021) further shows how the structure of EV subsidies interacts with local infrastructure, affecting both the supply and demand sides of the market. Similarly, Bushnell, Muehlegger, and Rapson (2022) examine how regional electricity prices and gasoline costs jointly influence consumer vehicle choice. Together, these studies emphasize that the adoption of EVs is shaped not only by vehicle features but also by broader spatial and infrastructural constraints.

This work also contributes to a growing literature on market structure and dynamic competition in the automobile industry. Grieco, Murry, and Yurukoglu (2024) use a logit demand model to analyze evolving market power and consumer welfare in the U.S. automotive sector from 1980 to 2018, while Hashmi and Biesebroeck (2016) study how competition and innovation interact over time in global vehicle markets. These studies provide a macro-level view of competition and welfare in differentiated product markets and establish the importance of modeling firm strategy and product quality jointly.

While this body of work has advanced our understanding of vehicle demand and market evolution, fewer studies have directly quantified how demographic characteristics—such as income, education, and urban residence—mediate consumer utility from EV-specific attributes like battery capacity and driving range. This paper extends the literature by using a structural demand model to recover heterogeneous preferences over EV features across population subgroups and by linking those preferences to observed demographic variation in adoption behavior.

The remainder of the paper is organized as follows. Section 2 provides industry context. Section 3 describes the data and presents summary statistics. Section 4 introduces the demand model. Section 5 reports the estimation results. Section 6 concludes.

2 Industry Background

The electric vehicle (EV) industry has undergone a remarkable transformation over the past two decades. In the early 2000s, EVs faced significant barriers to widespread adoption, including high production costs, limited battery capacity, and short driving ranges. Consumer

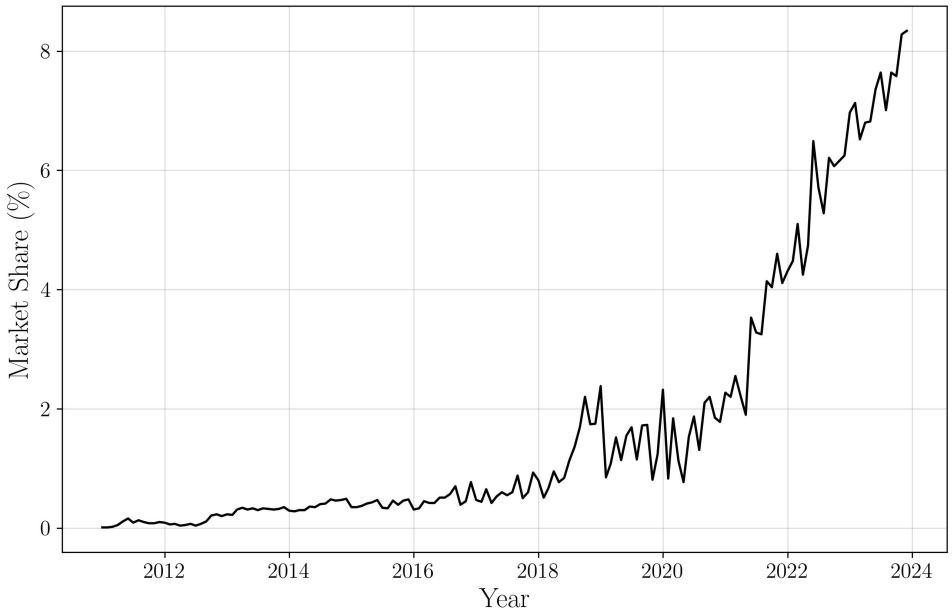


Figure 1: Electric Vehicle Share of New Vehicle Sales in the United States (2012–2024)

Notes: The annual share of battery electric vehicles (BEVs) in total new light-duty vehicle sales in the United States from 2012 to 2024. Data are from Ward's Automotive Yearbooks.

perceptions of EVs were shaped by these constraints, which relegated them to niche status despite growing environmental awareness. However, the introduction of lithium-ion battery technology gradually improved energy density and reduced costs, laying the foundation for broader market acceptance.

Between 2012 and 2023, the EV sector experienced rapid technological and commercial progress. Battery pack costs declined from over $1,000 per kilowatt-hour in 2010 to under $150 by 2023, while average driving range more than doubled, with many models now exceeding 250 miles on a single charge. These advancements made EVs increasingly competitive with internal combustion engine (ICE) vehicles, both in terms of performance and operational cost. These improvements have contributed to growing consumer acceptance and market penetration of EVs. As shown in Figure 1, the share of battery electric vehicles in new light-duty vehicle sales increased from less than 0.1% in 2012 to over 8% in 2024, reflecting a rapid transformation in vehicle purchase behavior.

The competitive landscape also evolved significantly. Tesla's early success demonstrated the viability of high-performance EVs and changed public expectations about the technology. The launch of the Nissan Leaf in 2010 marked one of the first mass-market, affordable EVs, helping to normalize electric driving for a broader audience. In response, traditional au-

tomakers—including General Motors, Volkswagen, and Ford—intensified their electrification strategies, expanding their EV portfolios. New entrants such as Rivian and Lucid Motors have also entered the market, targeting premium and utility segments, thereby increasing competition and innovation in the sector.

Government policy has played a central role in shaping the industry's growth trajectory. Financial incentives, including tax credits, rebates, and subsidies, have lowered the effective purchase price for EVs and stimulated consumer demand. Regulatory frameworks such as California's Zero Emission Vehicle (ZEV) mandate have pressured automakers to increase their share of clean vehicles. Looking ahead, ongoing investments in charging infrastructure, solid-state battery development, and vehicle-to-grid integration are expected to further enhance the appeal and functionality of EVs. Together, these technological and institutional trends suggest continued momentum toward widespread EV adoption in the coming decade.

3 Data

This study combines two data sources: market-level information on new vehicles and micro-level data on households that purchase them.

3.1 Automobile Market Data

The new vehicle data is sourced from Ward's Automotive Yearbooks, which provides detailed information on sales, manufacturer suggested retail prices (MSRP), and characteristics of new cars and light trucks sold in the United States from 2014 to 2024. Vehicle characteristics include a measure of acceleration (horsepower/curb weight), vehicle dimensions, drive type, fuel efficiency, and fuel type. For non-EVs, the data also include engine size, while for EVs, it reports battery capacity and driving range. To construct market shares, I define the market size as the total number of households in the United States divided by 2.5. This reflects the empirical observation that the average household owns nearly two cars, and the average tenure of car ownership is approximately five years, consistent with industry reports and previous academic studies.

Each automotive brand (make) produces multiple models, and each model may be available in several variants (trims). Although my dataset includes detailed specifications and prices at the individual trim level, sales data is available only at the model level. Consequently, similar to previous studies (see Berry, Levinsohn, and Pakes (1995), Berry, Levin-

Table 1: Summary Statistics for Electric and Non-Electric Vehicles (2014–2024)

	Non-Electric		Electric	
	Mean	Standard Deviation	Mean	Standard Deviation
MSRP (1000 USD)	27.24	9.43	37.20	13.82
Market Share (%)	0.40	0.37	0.27	0.26
Car	0.31	0.46	0.39	0.49
SUV	0.48	0.50	0.60	0.49
Truck	0.17	0.37	0.01	0.09
Van	0.04	0.19	0.00	0.03
AWD	0.75	0.43	0.78	0.42
Height (ft)	5.46	0.55	5.13	0.33
Length (ft)	15.94	1.27	15.41	0.75
Width (ft)	6.21	0.30	6.22	0.28
Curbweight (1000 lb)	3.76	0.70	4.23	0.66
Horsepower	223.96	69.65	218.86	61.20
MPG	26.06	5.56	116.45	14.53
Engine Size (L)	2.69	0.92	-	-
Hybrid	0.01	0.12	-	-
FlexFuel	0.09	0.29	-	-
Battery Capacity (kWh)	-	-	70.76	22.37
Driving Range (Mi)	-	-	271.70	61.84

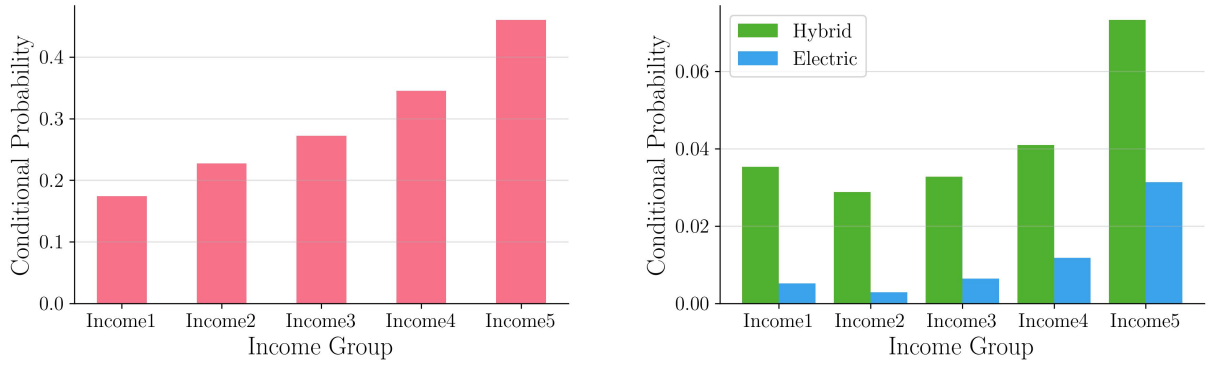
Notes: Data are from Ward's Automotive Yearbooks. Each observation corresponds to a make–model–year vehicle, using pricing and characteristics from the base trim of each model. Summary statistics are weighted by model-level sales. All prices are reported in 2015 U.S. dollars (thousands).

sohn, and Pakes (2004), Goldberg (1995), and Petrin (2002)), I use the base trim's characteristics and price to represent the model.

Table 1 presents summary statistics for the dataset, consisting of almost 3,000 observations at the make-model-year level, including 163 battery-electric vehicles. EVs typically have higher average prices (MSRP) and lower market shares than non-EVs. EVs also feature greater fuel efficiency, measured by miles per gallon equivalent (MPGe).

3.2 Consumer Choices and Demographics

Consumer information comes from the Consumer Expenditure Survey (CEX), a rotating panel that records U.S. household purchasing patterns. I use data from the years 2014 to 2024, which include nearly 230,000 household-level observations, to approximate the empirical distribution of demographic characteristics in the United States. In addition, I use the CEX automobile supplement, which links vehicle purchases to purchaser demographics.



(A) All New Vehicle Purchase

(B) Hybrid and EV Purchase

Figure 2: Conditional Probabilities of New Vehicle Purchase by Income Quantiles

Notes: Data are from the Consumer Expenditure Survey (CEX). Panel A shows the conditional probability of new vehicle purchases by income quintile. Panel B presents the conditional probability of purchasing a hybrid or electric vehicle, conditional on purchasing any new vehicle, across income quintiles.

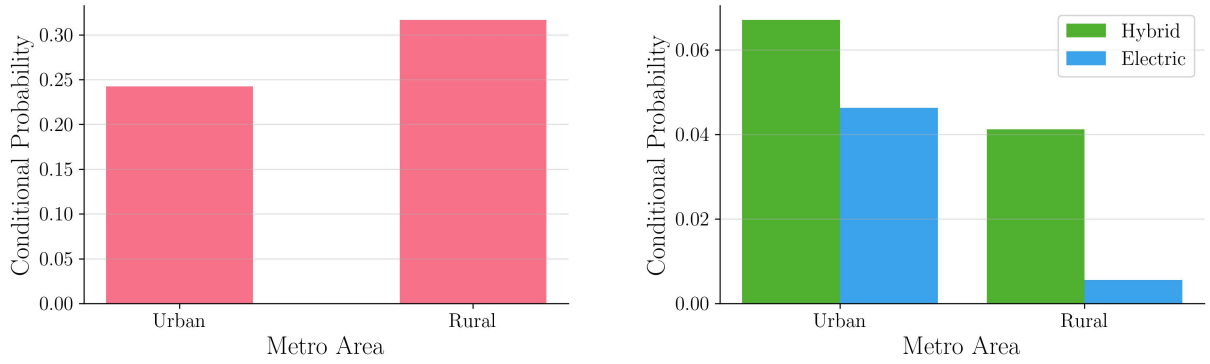
This supplement contains approximately 78,000 observations of new vehicle purchases.

I use these data to estimate the probability of new vehicle purchase by demographic group. I construct micro-moments from this data to use as targets for the heterogeneous agent demand model, following Goldberg (1995), Petrin (2002), and Berry, Levinsohn, and Pakes (2004). These moments capture conditional purchase probabilities by demographic group and are informative about the variation in consumer preferences.

Some patterns in the data help motivate specification choices for the demand model. For example, the observation that urban households are more likely to purchase EVs suggests stronger preferences for such vehicles among these groups compared to rural households.

To approximate the broader distribution of household demographics, I sample from the Current Population Survey (CPS) for the years 2014 to 2024, using the same demographic variables as those observed in the CEX. This allows for consistent integration of population-level characteristics into the demand estimation model.

Figure 2 displays the conditional probabilities of new vehicle purchase by income quantile. Panel A shows that the conditional probability of purchasing a new vehicle increases monotonically with income. Households in the highest income quintile are more than twice as likely to purchase a new vehicle compared to those in the lowest quintile. Panel B disaggregates new vehicle purchases into hybrid and electric vehicle purchase. While hybrid adoption is relatively stable across income groups, it rises sharply in the top quintile. EV



(A) All New Vehicle Purchase

(B) Hybrid and EV Purchase

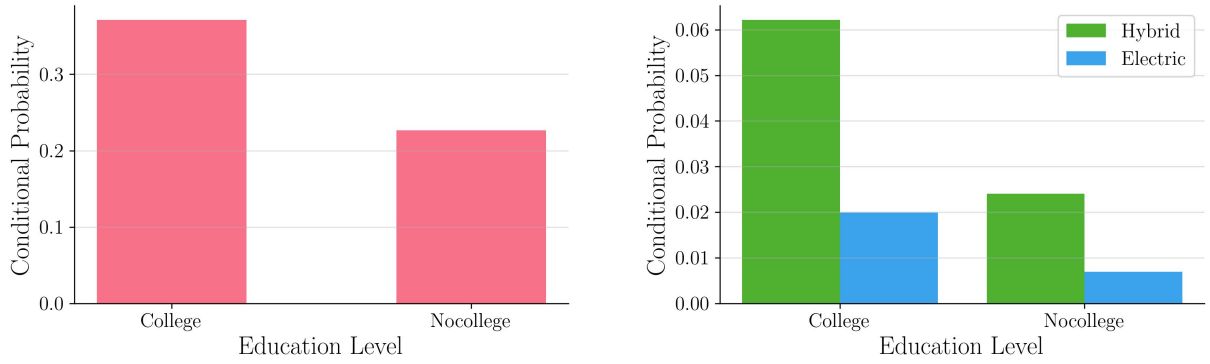
Figure 3: Conditional Probabilities of New Vehicle Purchase by Urban-Rural Areas

Notes: Data are from the Consumer Expenditure Survey (CEX). Panel A shows the conditional probability of new vehicle purchases by urban versus rural residence. Panel B shows the probability of purchasing a hybrid or electric vehicle, conditional on buying any new vehicle, by urban versus rural residence.

purchases, by contrast, exhibit a much steeper income gradient, with a notably higher probability in the highest income group. This pattern suggests that high-income households not only purchase more vehicles overall, but are disproportionately more likely to adopt EVs.

Figure 3 presents the conditional probabilities of new vehicle purchase by urban-rural areas. Panel A indicates that urban households are somewhat less likely to purchase new vehicles overall compared to rural households. However, the right panel reveals that urban households are significantly more inclined to adopt hybrid and electric vehicles. The probability of purchasing an electric vehicle in urban areas is nearly five times higher than in rural areas, while hybrid adoption is also noticeably greater among urban households. This urban-rural divide is especially pronounced for EVs, suggesting that infrastructure, availability, or preferences in urban environments may play a crucial role in facilitating clean vehicle adoption. The results underscore the heterogeneity in vehicle adoption not only across income levels (as seen in Figure 2), but also across geographic settings.

Figure 4 illustrates the conditional probabilities of new vehicle purchase by education levels. Panel A shows that college-educated households are substantially more likely to purchase new vehicles compared to households without a college degree. Panel B further decomposes these purchases into hybrid and electric vehicle purchase, revealing sharp educational disparities in clean vehicle adoption. College-educated households are more than twice as likely to purchase hybrids and over three times more likely to purchase EVs. The steeper gradient for EVs indicates that education plays a particularly strong role in the adoption of



(A) All New Vehicle Purchase

(B) Hybrid and EV Purchase

Figure 4: Conditional Probabilities of New Vehicle Purchase by Education Levels

Notes: Data are from the Consumer Expenditure Survey (CEX). Panel A presents the conditional probability of purchasing a new vehicle by educational attainment (college-educated vs. non-college). Panel B reports the probability of purchasing a hybrid or electric vehicle, conditional on buying a new vehicle, by education level.

emerging, environmentally friendly technologies.

To complement the graphical analysis, Table 2 reports summary statistics for demographic characteristics across the full U.S. population, EV purchasers, and hybrid purchasers. The table reveals stark compositional differences between adopters of clean vehicles and the broader population. EV purchasers have significantly higher average income ($160,662 vs. $67,422) and are more likely to live in urban areas (71% vs. 25%) and to hold a college degree (81% vs. 47%). These patterns underscore the demographic skew in clean vehicle adoption and highlight the importance of including income, education, and geography in the model of heterogeneous preferences. Column (4) further reports the probability of new vehicle purchase by subgroup, confirming that purchase incidence increases strongly with income and education.

4 Model

This section presents the structural demand model used to estimate consumer preferences for vehicle attributes and heterogeneity in willingness-to-pay (WTP) for electric vehicles (EVs). The framework follows the differentiated product demand model of Berry, Levinsohn, and Pakes (1995), extended to incorporate random coefficients, demographic interactions, and micro-moment conditions following Petrin (2002).

Table 2: Demographic Distribution of Hybrid and Electric Vehicle Purchases in the United States (2014–2024)

Demographic Groups	United States (1)	Purchasers of EVs (2)	Purchasers of Hybrids (3)	Purchase Probability (4)
Income	67.422	160.662	128.866	
Q1 Income	0.200	0.041	0.089	0.175
Q2 Income	0.200	0.030	0.094	0.228
Q3 Income	0.200	0.079	0.128	0.272
Q4 Income	0.200	0.187	0.203	0.345
Q5 Income	0.200	0.662	0.487	0.460
Urban	0.250	0.707	0.322	0.242
Rural	0.750	0.293	0.678	0.317
College	0.470	0.813	0.797	0.371
No College	0.530	0.187	0.203	0.227

Notes: Data are from the Consumer Expenditure Survey (CEX). The table reports average values of key demographic characteristics for the overall U.S. population (column 1), electric vehicle (EV) purchasers (column 2), hybrid vehicle purchasers (column 3), and new vehicle purchase probabilities by subgroup (column 4). Income is measured in thousands of 2015 U.S. dollars. Purchase probability refers to the share of households in each demographic group that purchased a new vehicle during the sample period.

Consumer utility. In each period t, consumer i makes a discrete choice among among a set of $\mathcal{J}_t$ differentiated vehicle models available in year t and an outside "nopurchase" option (indexed 0), choosing the option that delivers the maximum conditional indirect utility. The indirect utility of a consumer i from purchasing a vehicle model j during year t is given by:

$$u_{ijt} = \alpha_i p_{jt} + \beta_i \mathbf{x}_{jt} + \xi_{jt} + \epsilon_{ijt}, \tag{1}$$

where p_{jt} is the price of vehicle j, and α_i reflects income-dependent marginal utility of price, varying across individuals, $\mathbf{x}_{jt}$ is vector of observed vehicle characteristics (e.g., size, fuel economy, driving range), ξ_{jt} represents unobserved (to the econometrician) vehicle characteristics, and ϵ_{ijt} captures idiosyncratic consumer-vehicle-specific variation.

The consumers may decide not to purchase any of the vehicles. To account for this possibility, an outside good is included in the model, allowing consumer i the option not to buy any of the $\mathcal{J}_t$ vehicles available in the market. The mean utility of the outside good, δ_{0t}, is normalized to be constant over time and equal to 0.

Heterogeneity in preferences. We model consumer heterogeneity by interacting household demographics and unobserved preferences with car attributes. Our baseline specification

is:

$$\alpha_i = \bar{\alpha} + \sum_h \alpha_h \mathbf{D}_i^h$$

$$\beta_{ik} = \bar{\beta}_k + \sum_h \beta_{kh} \mathbf{D}_i^h + \sigma_k v_{ik},$$

where subscript k denotes the k th car characteristic (including a constant) and h indexes dimensions of consumer demographics (e.g., income). Allowing for observed heterogeneity allows substitution patterns to differ by demographics. The distribution of D_i is taken from the CPS. In practice, we do not interact every demographic with every car characteristic. Allowing for unobserved heterogeneity allows for more flexible substitution patterns. Unobserved taste for automobile characteristics, v_{ik}, are assumed to be independent draws from the standard normal distribution.

Marker shares. For a given year, market shares in the model are given by integrating over the distribution of consumers who vary in their demographics, unobserved tastes for characteristics, and idiosyncratic error terms,

$$s_{jt} = \int_i \frac{\exp\left(-\alpha_i p_{jt} + \beta_i \mathbf{x}_{jt} + \xi_{jt}\right)}{1 + \sum_{l \in \mathcal{J}_t} \exp\left(-\alpha_i p_{lt} + \beta_i \mathbf{x}_{lt} + \xi_{lt}\right)} dF(i) \tag{2}$$

Estimation strategy. The model is estimated using the generalized method of moments (GMM), following the inversion procedure outlined by Berry, Levinsohn, and Pakes (1995). Price is instrumented using characteristics of competing products (BLP instruments). In addition to matching observed market shares, I include micro-moment conditions that match the model-predicted demographic profiles of vehicle adopters to their empirical counterparts in the CEX. These moments improve the identification of heterogeneous preferences and allow the model to separately recover differences in valuation by income, education, and geography.

5 Empirical Results

Table 3 reports the estimated demand coefficients from three model specifications: Ordinary Least Squares (OLS) Logit, Instrumental Variables (IV) Logit, and a Random Coefficients Logit model estimated with instrumental variables and micro-moment conditions. All specifications include make and year fixed effects. The strength of the instruments is reflected in a first-stage F-statistic of 13.2, indicating that weak instrument concerns are unlikely to

Table 3: Results

Variable	OLS Logit (1)	IV Logit (2)	Random Coefficients (3)
	A. Price Coefficients (α's)		
α_1	-2.165^{***}	-2.965^{***}	-7.944^{***}
α_2			-6.422^{**}
α_3			-5.150^{**}
α_4			-3.528^{**}
α_5			-1.592^{**}
	B. Base Coefficients (β's)		
Space	0.840^{***}	0.971^{***}	1.072^{***}
Horsepower/Weight	0.212^{*}	0.340^{**}	0.413^{***}
MPG	0.979^{***}	0.861^{***}	1.018^{***}
AWD	0.356^{**}	0.372^{**}	0.512^{***}
SUV	0.304^{*}	0.209	0.316^{**}
Truck	-0.174	-0.530	-0.544
Van	-0.629^{*}	-0.813^{*}	-0.832^{***}
EV	-10.280^{***}	-10.040^{***}	-14.566^{***}
Hybrid	-2.391^{***}	-2.000^{***}	-3.149^{***}
Battery Capacity	0.442^{**}	0.425^{***}	0.299^{***}
Driving Range	1.021^{*}	1.142^{**}	1.228^{***}

Notes: Unit of observation is year–make–model, from 2014 to 2024. Standard errors are clustered at the make level. All continuous vehicle attributes are log-transformed and standardized. Price is measured in 2015 U.S. dollars per $10,000.
* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

be severe.

Price sensitivity. Panel A of Table 3 reports the estimated price coefficients. In columns (1) and (2), the price coefficient is negative and statistically significant. The stronger magnitude under IV reflects the correction for price endogeneity, which tends to bias OLS estimates toward zero.

Column (3) presents results from the random coefficients model, where price sensitivity varies by income quintile. This model allows for consumer heterogeneity in price sensitivity, and yields substantially larger (more negative) coefficients. The resulting coefficients exhibit a U-shaped pattern: both the lowest- and highest-income groups are less price-sensitive, while middle-income households display steeper demand slopes. This pattern is visualized in Table 5, which shows the estimated marginal utility of price (α) across income groups.

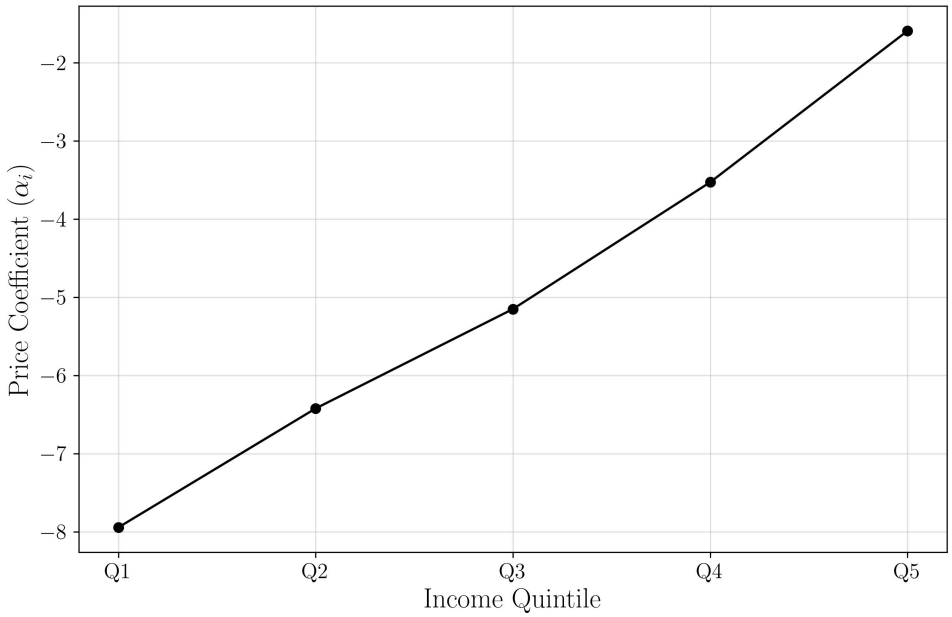


Figure 5: Estimated Price Coefficient (α) by Income

Notes: Horizontal axis shows income quintiles (Q1: lowest; Q5: highest). Vertical axis reports the estimated price coefficient (α), where more negative values indicate greater price sensitivity.

Attribute preferences. Panel B of Table 3 presents estimates of consumer preferences for vehicle characteristics. Across all specifications, consumers exhibit strong positive preferences for vehicle space, fuel efficiency (measured by miles per gallon), all-wheel drive (AWD), and acceleration (proxied by horsepower-to-weight ratio), with magnitudes increasing in the random coefficients model.

The coefficient on electric vehicles (EVs) and hybrid electric vehicles (HEVs) are large and negative across specifications, indicating a substantial baseline disutilities relative to conventional gasoline vehicles, especially in the case of EVs. In the random coefficients logit model, the EV and hybrid dummy coefficients estimated at -14.5 and -3.1, respectively. However, attributes unique to EVs—particularly battery capacity and driving range—are positively valued, suggesting that improvements in these features can partially offset the disutility of EV adoption.

Random coefficients. To explore variation in preferences across population segments, I estimate a model with demographic interactions, using the lowest-income, rural, non-college-educated group as the baseline. Table 4 reports the estimated taste coefficients for EVs and hybrids across income, education, and geography. The specification includes interaction

Table 4: Random Coefficient Estimation Results

Variable	EV (1)	HEV (2)
	A. Baseline (Q1 Income, Rural, Non-College)	
β	-14.566^{***}	-3.149^{**}
	B. Income Adjustments	
Q2 Income Adjustments	-1.508^{***}	-0.709^{***}
Q3 Income Adjustments	-1.097^{***}	-0.914^{***}
Q4 Income Adjustments	-0.676	-1.012^{***}
Q5 Income Adjustments	1.050^{***}	-0.652^{*}
	C. Urban/College Adjustments	
Urban Adjustments	4.985^{***}	1.693^{***}
College Adjustments	3.139^{***}	2.226^{***}

Notes: Unit of observation is year–make–model, from 2014 to 2024. Standard errors are clustered at the make level. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

terms that shift baseline utility according to demographic characteristics. The coefficients for income quintiles (Q2–Q5) reflect how utility differs from the baseline as income increases. For EVs, income adjustments are increasingly less negative and become positive for the highest income group: Q2 adds −1.5, Q3 adds −1.0, Q4 adds −0.6, and Q5 adds +1.0. This implies that the total utility of EVs rises from −14.5 in Q1 to approximately −16.0, −15.5, −15.1, and −13.5 in Q2 through Q5, respectively. Figure 6 visualises this pattern. For hybrids, Q2 through Q5 adjust the baseline by −0.7, −0.9, −1.0, and −0.6, respectively. Figure 7 visualises this pattern.

Urban and college-educated households exhibit significantly higher utility from EV and hybrid ownership. For EVs, living in an urban area increases utility by +4.0, and having a college degree adds +3.1. These effects are large relative to the baseline disutility and suggest that urban, college-educated households experience much less resistance to EV adoption. For hybrids, the urban and college adjustments are smaller (+1.6 and +2.2, respectively), but still significant and positive.

Taken together, the results imply that EV adoption is highly segmented across demographics: while higher-income, urban, and college-educated households have more favorable views of EVs, the baseline disutility remains large for disadvantaged populations. In contrast, hybrids face a smaller baseline penalty and exhibit more uniform appeal across income groups, but higher appealing among urban and college-educated households. These find-

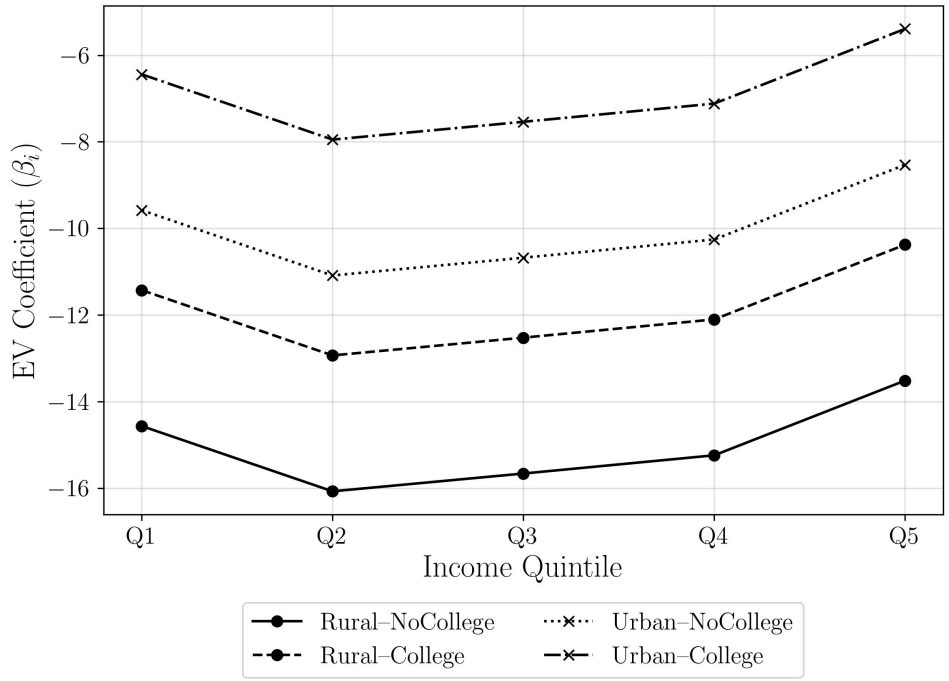


Figure 6: Estimated EV Taste Coefficient (β) by Income × College × Urban

Notes: The horizontal axis denotes income quintiles from lowest (Q1) to highest (Q5). The vertical axis reports the taste coefficient for EVs, where higher values indicate stronger preferences.

ings underscore the importance of demographic targeting in electric vehicle policy design, particularly in mitigating the fixed disutility of EV adoption among rural and low-income populations.

Willingness to pay. Finally, I compute implied willingness-to-pay (WTP) estimates for EVs and hybrids by demographic group. WTP for characteristic k and demographic group i is calculated as the ratio of the group-specific taste coefficient (β_{ik}) to the marginal utility of price (α_i), $-\beta_{ik}/\alpha_i$. Figures 8 and 9 report the WTP of EVs and hybrids values across income quintiles, stratified by education and urban-rural areas.

Figure 8 reports estimated WTP for EVs across different subgroups. The figure reveals a pronounced downward trend in WTP as income increases, implying that higher-income households require larger incentives to adopt EVs. This pattern is most striking for rural, non-college-educated households, whose WTP declines sharply with income, reaching the lowest values among all groups in the top quintile. In contrast, urban, college-educated households maintain relatively higher WTP across all income levels, indicating stronger

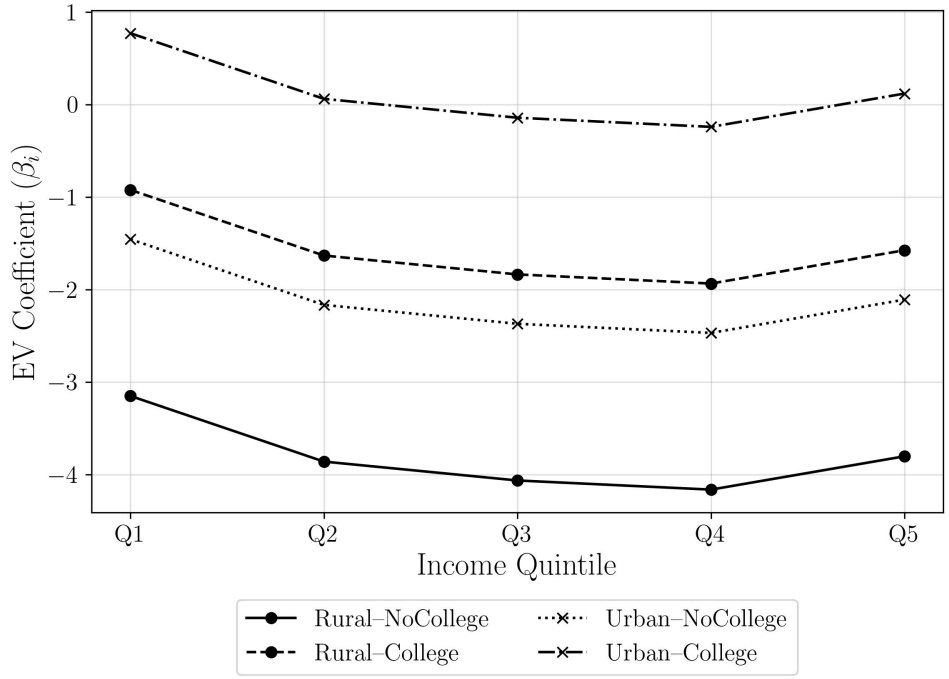


Figure 7: Estimated Hybrid Vehicle Taste Coefficient (β) by Income × College × Urban

Notes: The horizontal axis denotes income quintiles from lowest (Q1) to highest (Q5). The vertical axis reports the taste coefficient for hybrid vehicles, where higher values indicate stronger preferences.

preferences for EVs. The figure highlights significant heterogeneity in EV valuation, with educational attainment and urban residency substantially mitigating the disutility associated with EV adoption. These results underscore the need for targeted policy interventions that account for both price sensitivity and fixed disutility. EV subsidies may be most effective when directed toward middle-income or rural populations.

Figure 9 depicts WTP for hybrids by the same demographic dimensions. While the overall structure mirrors that of Figure 8, key differences emerge. The disutility of hybrids is smaller across all groups, reflected in the less negative WTP values compared to EVs. The income gradient is also flatter, especially among urban and college-educated households, suggesting that hybrids face more uniform acceptance across socioeconomic segments. Although rural, non-college-educated households still show the lowest WTP, the gap relative to other groups is narrower. These results imply that hybrids may serve as a more universally acceptable transitional technology and highlight the nuanced role that demographic factors play in shaping preferences for alternative fuel vehicles.

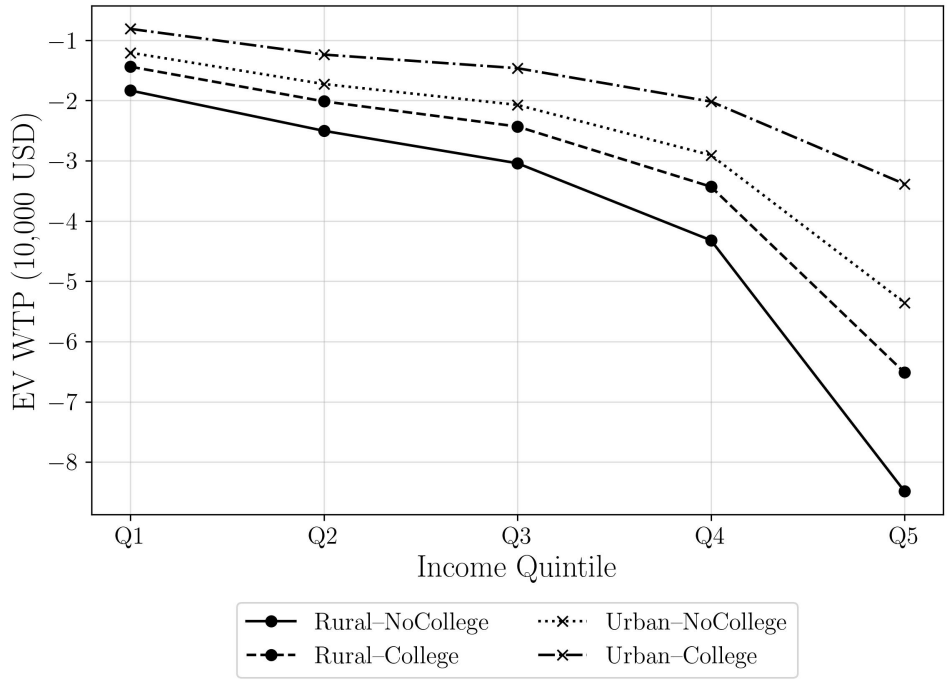


Figure 8: Estimated EV WTP by Income × College × Urban

Notes: Horizontal axis denotes income quintiles (Q1–Q5). The vertical axis reports estimated willingness-to-pay (WTP) for EVs, derived as the ratio of taste coefficient (β) to price sensitivity (α). Higher values indicate greater willingness to adopt the vehicle type.

6 Conclusion

This paper quantifies the determinants of electric vehicle demand and highlights the pronounced heterogeneity in consumer preferences that underlie adoption behavior in the U.S. automotive market. By estimating a random coefficients logit model using rich vehicle and consumer data, I find that while improvements in driving range and battery capacity raise the perceived value of EVs, baseline utility from EV ownership remains significantly negative for key demographic segments—particularly rural, low-income, and non-college-educated consumers.

Importantly, the results show that income, education, and urban residence jointly mediate consumer response to EV characteristics and pricing. Urban, college-educated consumers are considerably more receptive to EVs and exhibit higher willingness-to-pay, even controlling for price sensitivity. In contrast, high-income consumers require disproportionately large price reductions to induce adoption, despite their greater ability to pay. These patterns imply that a one-size-fits-all policy approach may be inefficient. Instead, targeted

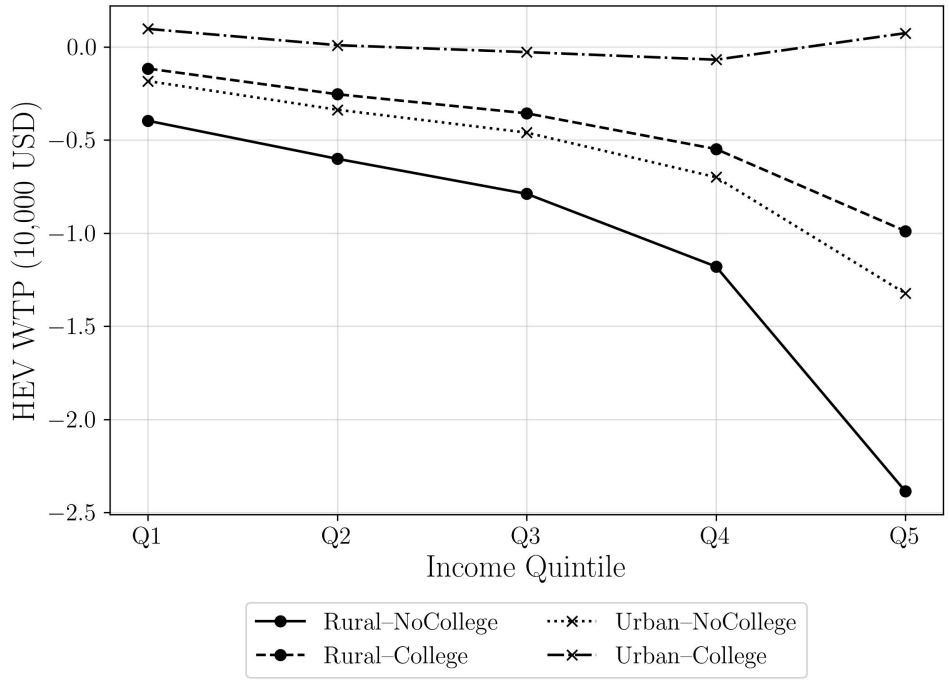


Figure 9: Estimated hybrids WTP by Income × College × Urban

Notes: Horizontal axis denotes income quintiles (Q1–Q5). The vertical axis reports estimated willingness-to-pay (WTP) for hybrids, derived as the ratio of taste coefficient (β) to price sensitivity (α). Higher values indicate greater willingness to adopt the vehicle type.

interventions—such as subsidies for disadvantaged groups, investments in rural charging infrastructure, and education campaigns—may be more effective in expanding the EV market.

References

Berry, Steven, James Levinsohn, and Ariel Pakes (1995). "Automobile Prices in Market Equilibrium". In: *Econometrica: Journal of the Econometric Society*, pp. 841–890.

— (2004). "Differentiated products demand systems from a combination of micro and macro data: The new car market". In: *Journal of political Economy* 112.1, pp. 68–105.

Bushnell, James B, Erich Muehlegger, and David S Rapson (2022). *Energy prices and electric vehicle adoption*. Tech. rep. National Bureau of Economic Research.

Goldberg, Pinelopi Koujianou (1995). "Product differentiation and oligopoly in international markets: The case of the US automobile industry". In: *Econometrica: Journal of the Econometric Society*, pp. 891–951.

Grieco, Paul LE, Charles Murry, and Ali Yurukoglu (2024). "The evolution of market power in the us automobile industry". In: *The Quarterly Journal of Economics* 139.2, pp. 1201–1253.

Grigolon, Laura, Mathias Reynaert, and Frank Verboven (2018). "Consumer valuation of fuel costs and tax policy: Evidence from the European car market". In: *American Economic Journal: Economic Policy* 10.3, pp. 193–225.

Hashmi, Aamir Rafique and Johannes Van Biesebroeck (2016). "The relationship between market structure and innovation in industry equilibrium: a case study of the global automobile industry". In: *Review of Economics and Statistics* 98.1, pp. 192–208.

Li, Shanjun et al. (2017). "The market for electric vehicles: indirect network effects and policy design". In: *Journal of the Association of Environmental and Resource Economists* 4.1, pp. 89–133.

Petrin, Amil (2002). "Quantifying the benefits of new products: The case of the minivan". In: *Journal of political Economy* 110.4, pp. 705–729.

Springel, Katalin (2021). "Network externality and subsidy structure in two-sided markets: Evidence from electric vehicle incentives". In: *American Economic Journal: Economic Policy* 13.4, pp. 393–432.